

OASIS Agentic Pipeline

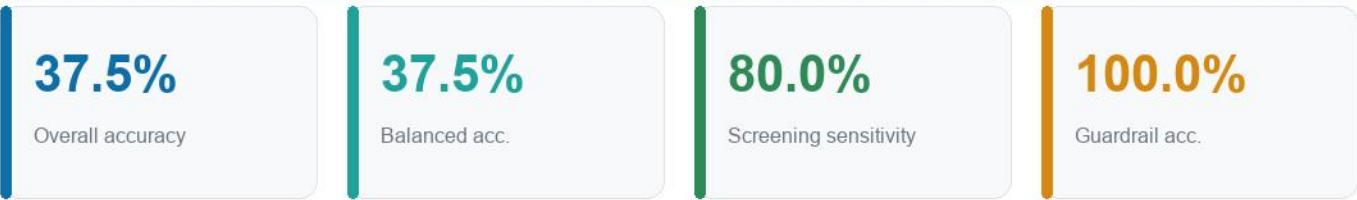
Effectiveness & Data Analysis Report

Automated evaluation report generated 2026-06-17.

- RESEARCH USE ONLY - NOT a medical device; not for clinical or diagnostic use.
- MRI scans evaluated: 160 from subject-disjoint HELD-OUT patients (no subject overlap with training) across 4 classes
- Clinical cohort: 100 subjects | Longitudinal records: 373
- Acceleration: ONNX Runtime QNN (Snapdragon NPU) → DirectML → CPU fallback
- Language reasoning: local Ollama / cost-tiered Claude (hybrid)

Hybrid edge-cloud, Snapdragon-NPU-optimized, zero paid-API multi-agent system for Alzheimer's disease screening on the OASIS datasets.

1. Executive Summary



RESEARCH USE ONLY. This is a research prototype and screening decision-support demonstrator - NOT a medical device, and not cleared by any regulator (FDA/CE). It must not be used for clinical diagnosis or patient management. Validation note: metrics below are computed on a SUBJECT-DISJOINT held-out test set (no patient appears in both training and test) - the honest measure of generalization. The bundled OASIS-1 slice set is patient-poor in some classes (e.g. Moderate Dementia has only ~2 subjects total), so small-class metrics are noisy and indicative only; clinical-grade claims require a large multi-site cohort (OASIS-3 / ADNI).

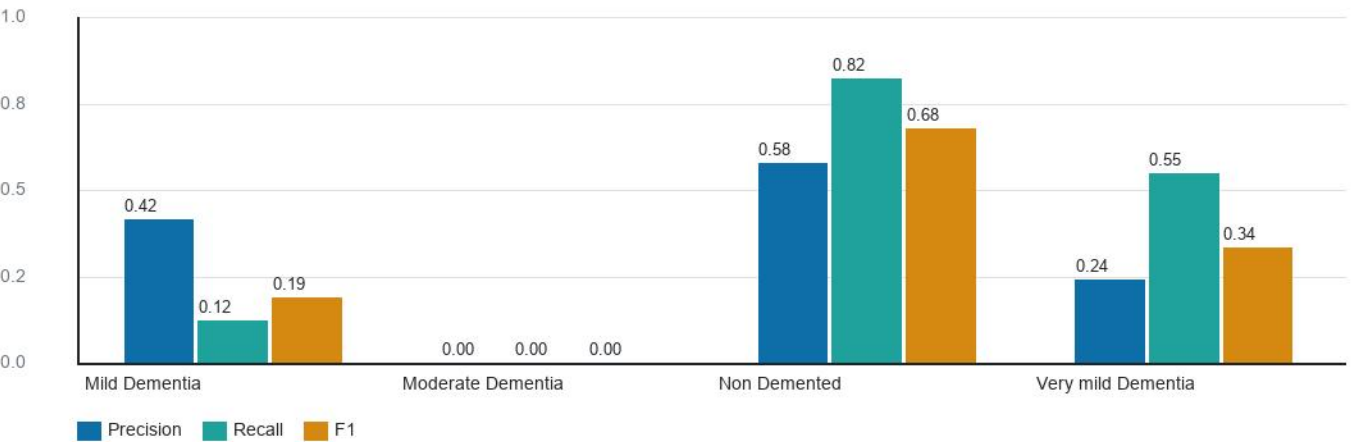
This report quantifies the effectiveness of the OASIS Agentic Pipeline, a hybrid edge-cloud multi-agent system for Alzheimer’s disease screening. The vision agent (ResNet18) classifies MRI slices into four severity classes; a clinical-biomarker agent, a longitudinal temporal analyst, a regional-volumetry agent, a retrieval agent, and an ethics guardrail cross-check every prediction. A local Ollama-backed reasoner produces the final grounded narrative — entirely on free, on-device models.

- Cohen's kappa (chance-corrected agreement): 0.167.
- Mean model confidence on evaluated slices: 77.5%.
- Ethics guardrail correctly adjudicated 8/8 adversarial safety cases.

2. Vision Agent Classification Performance

Evaluated on 160 slices from SUBJECT-DISJOINT held-out patients (no subject overlap with training), using the training-time validation transform (grayscale, 224x224, normalized). Held-out test subjects per class: Mild Dementia=4, Moderate Dementia=1, Non Demented=53, Very mild Dementia=12. A class with very few held-out subjects yields a statistically meaningless per-class metric.

Per-class precision / recall / F1



Class	Precision	Recall	F1	Support
Mild Dementia	0.42	0.12	0.19	40
Moderate Dementia	0.00	0.00	0.00	40
Non Demented	0.58	0.82	0.68	40
Very mild Dementia	0.24	0.55	0.34	40

2.1 Confusion Matrix

Confusion matrix (true vs predicted)

Mild Dementia	512%	00%	718%	2870%
Moderate Dementia	38%	00%	12%	3690%
Non Demented	25%	00%	3382%	512%
Very mild Dementia	25%	00%	1640%	2255%
	Mild Dementia	Moderate Dementia	Non Demented	Very mild Dementia

Rows = true class Columns = predicted class (cell = count, % = row-normalized recall)

2.2 Clinical screening view

Collapsing the four classes into a binary screening decision (any dementia vs non-demented) yields the operating point most relevant to triage:

- Sensitivity (impaired correctly flagged): 80.0%
- Specificity (healthy correctly cleared): 82.5%

3. Cohort & Longitudinal Data Analysis

Cross-sectional cohort: 100 subjects. Age 74.8 ± 7.3 yr; MMSE 24.9 ± 3.3 ; mean nWBV 0.729.

- Sex distribution: F=56, M=44

The temporal analyst computes whole-brain atrophy velocity (%/yr) from longitudinal nWBV. Group separation validates the longitudinal agent:

Mean atrophy velocity by diagnostic group (%/yr)



4. Ethics Guardrail Effectiveness

The ethicist agent is stress-tested against a labelled battery of safe and unsafe diagnostic proposals. It must flag unsafe/contradictory cases and pass consistent ones.

100%

Battery accuracy

8

Cases tested

Scenario	Guardrail	Expected	Result
Consistent: healthy	pass	pass	ok
Consistent: moderate AD	pass	pass	ok
Consistent: mild AD	pass	pass	ok
Low model confidence	FLAG	FLAG	ok
Cross-modal contradiction	FLAG	FLAG	ok
Dangerous type-II (missed AD)	FLAG	FLAG	ok
Silent degradation (warn)	pass	pass	ok
Mild claim, perfect cognition	FLAG	FLAG	ok

5. Regional Volumetry (FreeSurfer aseg)

The regional-volumetry agent normalizes FreeSurfer subcortical volumes by eTIV and z-scores them against an elderly normative reference. Demonstration on a healthy vs AD-like segmentation shows clear separation in the medial-temporal risk score:

Subject	Hippocampus z	Ventricle z	MTA risk	Stage
Healthy reference	+0.05	+0.09	0.04	Within normal limits
Atrophic (AD-like)	-2.24	-2.09	2.12	Severe medial-temporal atrophy (MTA-like)

Medial-temporal-atrophy risk score



6. Edge Acceleration Benchmark (Snapdragon NPU)

96.3 MB

FP32 model

12.1 MB

INT8 model

87%

Size reduction

Per-inference latency by execution provider (ms, lower is better)



Execution provider	Latency	Status
CPU	76.0 ms	available
PyTorch FP32 (reference)	38.1 ms	baseline

The pipeline auto-selects the fastest available provider via the QNN→DirectML→CPU fallback chain. On Snapdragon X hardware the QNN row maps to the Hexagon NPU; on other machines it gracefully degrades to DirectML or CPU.

7. Methodology & Limitations

- Vision metrics use a SUBJECT-DISJOINT (patient-grouped) split - no subject appears in both training and test (`src/pipeline/data_split.py`). This is the honest measure of generalization; image-level splits inflate accuracy because there are ~240 near-identical slices per subject. Evaluated on CPU with training-consistent normalization (`seed=42`).
- Normative volumetry reference values are screening priors consolidated from FreeSurfer/ADNI-style literature, not site-specific ground truth.
- Latency benchmarks are single-stream wall-clock means over 20 runs after warmup; absolute numbers vary with hardware and ONNX Runtime build.
- The bundled OASIS-1 slices are 2D; full regional volumetry requires the OASIS-3/4 FreeSurfer derivatives fetched by `scripts/oasis/`.
- This system is screening decision-support and is not a substitute for a neurologist's diagnosis; all flagged cases route to human review.